

Practical 9 - Estimating causal effects in trials

Survival Analysis

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: IPASS_IPD, EXCEL_IPD, ASCOT_IPD

R packages required: survival, survRM2

```
library(survival)
library(survRM2)
```

Introduction

For this practical you will need to use 3 datasets. Each of these datasets is an approximation of data from real trials created by digitizing the Kaplan-Meier curves of each study. Datasets are stored in .csv format. Part A uses the IPASS_IPD data, Part B uses the EXCEL_IPD data and Part C uses the ASCOT_IPD data.

Part A: Questions on the IPASS data

IPASS_IPD contains data approximating the IPASS trial referenced in the lecture notes and includes the following variables:

Variable	Description
patientID	unique patient identifier
months	time from randomization in months
event	1=cancer progression or death, 0=censored
treat	1=gefitinib, 0= carboplatin plus paclitaxel

Variable	Description
egfr	0=patient does not have EGFR mutation, 1=patient has EGFR mutation, 2= patient's EGFR status is unknown

Load the data:

```
ipass<-read.table("datasets/IPASS_IPD.csv",sep=" ",header=T)
```

Aims

Key learning objectives:

- Understand how to quantify treatment benefit using restricted mean survival time (RMST) and risk differences at specific time points.
- Understand how to assess the proportional hazards assumption and how it influences choice of summary measure.
- Encourage thinking about how the time horizon (τ) affects RMST and risk difference estimates.

Question 1

- Estimate the RMST and risk difference at 16 months using the code below.
- Write down what each means in plain English.
- Which do you prefer as a summary of treatment benefit?

```
#Kaplan-Meier by treatment
km<-survfit(Surv(months,event)~treat,data=ipass)
plot(km,ylab="Survival probability",
     xlab="Time since randomisation (months)",
     lwd=2,lty=c(1,2))
legend("topright",c("carboplatin + paclitaxel","gefitinib"),
     lwd=2,lty=c(1,2))

#RMST at 16 months
print(km,rmean=16)

#RMST (and their differences) using rmst2 function
rmst2(ipass$months, ipass$event, ipass$treat, tau=16)

#risks and risk diff at 16 months
```

```

risk.0<-1-summary(km,times=c(16))$surv[1]
risk.1<-1-summary(km,times=c(16))$surv[2]

risk.1-risk.0

#variance of risk diff, and 95% CI
varrisk.0<-summary(km,times=c(16))$std.err[1]^2
varrisk.1<-summary(km,times=c(16))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0

lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)

lower.ci.riskdiff
upper.ci.riskdiff

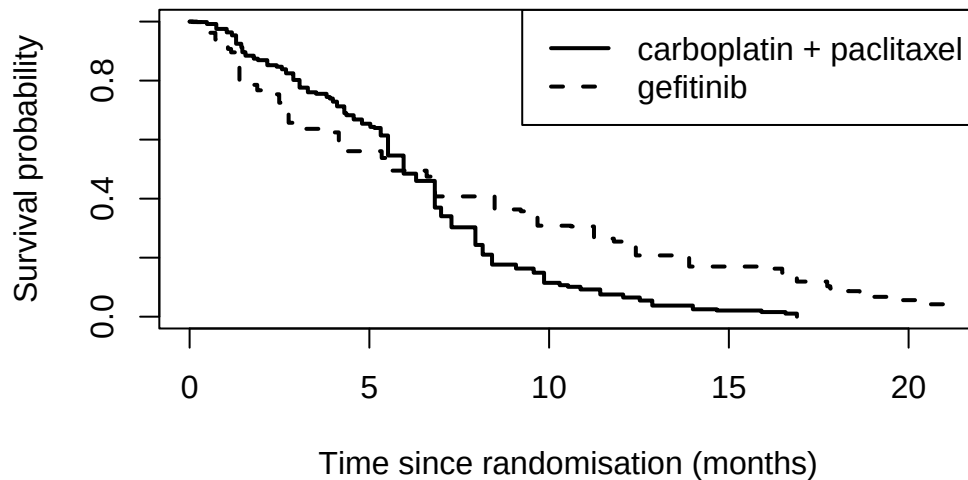
```

SOLUTIONS

```

#Kaplan-Meier by treatment
km<-survfit(Surv(months,event)~treat,data=ipass)
plot(km,ylab="Survival probability",
      xlab="Time since randomisation (months)",
      lwd=2,lty=c(1,2))
legend("topright",c("carboplatin + paclitaxel","gefitinib"),
      lwd=2,lty=c(1,2))

```



```
#RMST at 16 months
print(km, rmean=16)
```

```
Call: survfit(formula = Surv(months, event) ~ treat, data = ipass)
```

	n	events	rmean*	se(rmean)	median	0.95LCL	0.95UCL
treat=0	608	505	6.22	0.148	5.95	5.95	6.82
treat=1	608	448	7.22	0.231	5.60	5.34	6.83

* restricted mean with upper limit = 16

```
#RMST (and their differences) using rmst2 function
rmst2(ipass$months, ipass$event, ipass$treat, tau=16)
```

The truncation time: tau = 16 was specified.

Restricted Mean Survival Time (RMST) by arm

	Est.	se	lower .95	upper .95
RMST (arm=1)	7.222	0.231	6.770	7.674
RMST (arm=0)	6.215	0.148	5.926	6.505

Restricted Mean Time Lost (RMTL) by arm

	Est.	se	lower .95	upper .95
RMTL (arm=1)	8.778	0.231	8.326	9.230
RMTL (arm=0)	9.785	0.148	9.495	10.074

Between-group contrast

	Est.	lower .95	upper .95	p
RMST (arm=1)-(arm=0)	1.007	0.470	1.544	0
RMST (arm=1)/(arm=0)	1.162	1.075	1.256	0
RMTL (arm=1)/(arm=0)	0.897	0.845	0.952	0

```
#risks and risk diff at 16 months
risk.0<-1-summary(km,times=c(16))$surv[1]
risk.1<-1-summary(km,times=c(16))$surv[2]

risk.1-risk.0
```

```
[1] -0.154206
```

```
#variance of risk diff, and 95% CI
varrisk.0<-summary(km,times=c(16))$std.err[1]^2
varrisk.1<-summary(km,times=c(16))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0

lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)

lower.ci.riskdiff
```

```
[1] -0.1968636
```

```
upper.ci.riskdiff
```

```
[1] -0.1115484
```

- RMST difference at 16 months: 1.0 months (95% CI, 0.5-1.5) $p < 0.001$. Within 16 months of initiating therapy, time to either cancer progression or death is delayed by on average one month in the gefitinib group, compared to control.
- Risk difference: -15.4% (95% CI, -11.1% to -19.7%), $p < 0.001$. By 16 months, if 100 patients were to receive gefitinib instead of control, 15 fewer would have had either cancer progression or died by 16 months.
- Both estimates are meaningful. Which you prefer may be individual and there is no single correct answer. It could be argued that the RMST difference more clearly demonstrates that the treatment is beneficial here, because of the worse survival in the gefitinib group in the early part of follow-up, which we see in the Kaplan-Meier plot. Using both estimates together is also helpful, but cannot usually be done as a primary analysis!

Question 2

- For EGFR mutation-positive patients only (i.e. restrict your analysis to these patients) estimate the hazard ratio comparing gefitinib to carboplatin + paclitaxel, using a Cox proportional hazards model.
- Does the proportional hazards assumption look reasonable within this subgroup of patients?

```
#cox model
cox.mod<-coxph(Surv(months,event)~treat,data=ipass[ipass$egfr==1,])
summary(cox.mod)

#take a look at Kaplan-Meier plot
km<-survfit(Surv(months,event)~treat,data=ipass[ipass$egfr==1,])
plot(km,ylab="Survival probability",
     xlab="Time since randomisation (months)",
     lwd=2,lty=c(1,2))
legend("topright",c("carboplatin + paclitaxel","gefitinib"),
     lwd=2,lty=c(1,2))

#take a look at schoenfeld residuals
sch.resid<-cox.zph(cox.mod)
plot(sch.resid)
```

SOLUTIONS

```
#cox model
cox.mod<-coxph(Surv(months,event)~treat,data=ipass[ipass$egfr==1,])
summary(cox.mod)
```

Call:

```
coxph(formula = Surv(months, event) ~ treat, data = ipass[ipass$egfr ==
  1, ])
```

n= 261, number of events= 195

	coef	exp(coef)	se(coef)	z	Pr(> z)
treat	-0.6616	0.5160	0.1490	-4.441	8.95e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
treat	0.516	1.938	0.3854	0.691

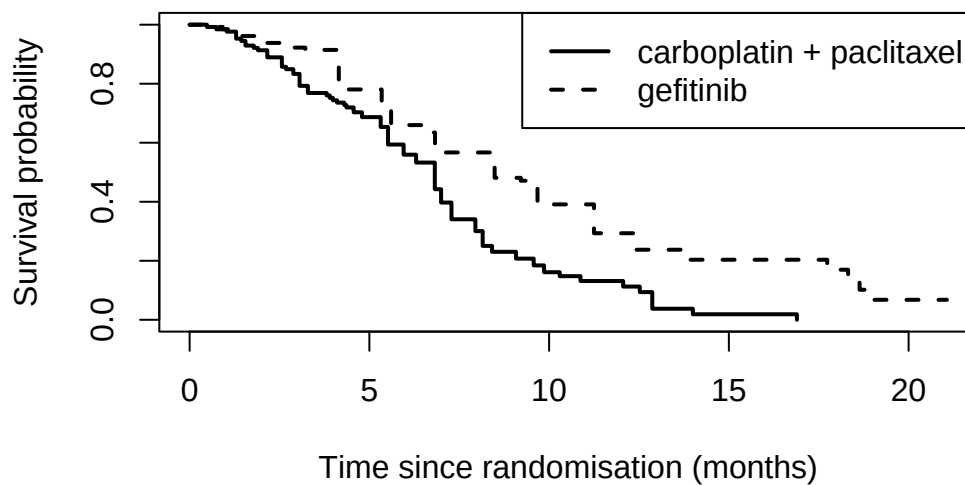
Concordance= 0.573 (se = 0.02)

Likelihood ratio test= 19.81 on 1 df, p=9e-06

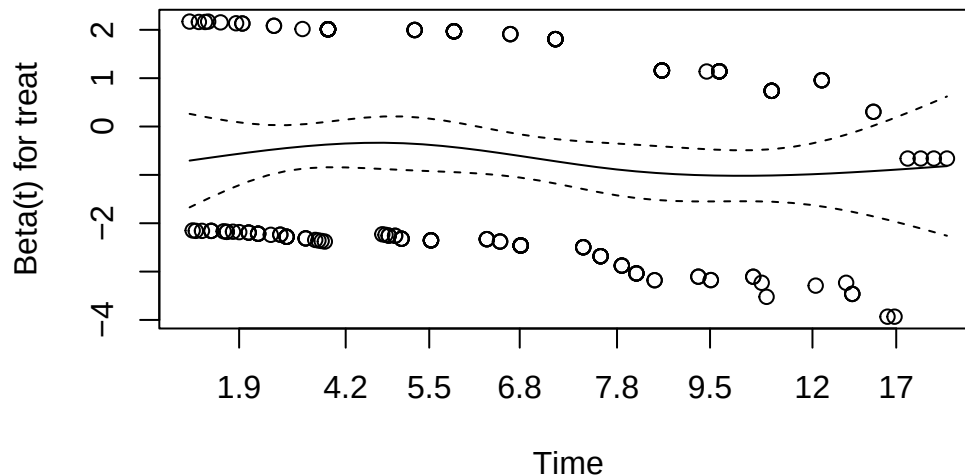
Wald test = 19.72 on 1 df, p=9e-06

Score (logrank) test = 20.36 on 1 df, p=6e-06

```
#take a look at Kaplan-Meier plot
km<-survfit(Surv(months,event)~treat,data=ipass[ipass$egfr==1,])
plot(km,ylab="Survival probability",
     xlab="Time since randomisation (months)",
     lwd=2,lty=c(1,2))
legend("topright",c("carboplatin + paclitaxel","gefitinib"),
     lwd=2,lty=c(1,2))
```



```
#take a look at schoenfeld residuals
sch.resid<-cox.zph(cox.mod)
plot(sch.resid)
```



- The hazard ratio estimate is 0.52 (95% CI 0.39-0.69, $p < 0.001$)
- The proportional hazards assumption looks far more reasonable within this subset of patients.

Question 3

- For EGFR mutation positive patients, estimate the risk difference and RMST at 8 months. Which is the most statistically significant? Why?
- Repeat the process for patients at 16 months and comment on your findings.

```
#RMST at 8 months
print(km,rmean=8)

#RMST (and their differences) using rmst2 function
rmst2(ipass$months[ipass$egfr==1], ipass$event[ipass$egfr==1],
      ipass$treat[ipass$egfr==1], tau=8)

#risk diffs at 8m
risk.0<-1-summary(km,times=c(8))$surv[1]
risk.1<-1-summary(km,times=c(8))$surv[2]

varrisk.0<-summary(km,times=c(8))$std.err[1]^2
varrisk.1<-summary(km,times=c(8))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0

lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)
```



```

risk.1-risk.0
lower.ci.riskdiff
upper.ci.riskdiff
2*pnorm((risk.1-risk.0)/sqrt(var.riskdiff))#p-value

```

SOLUTIONS

```

#-----
#RMST at 8 months
print(km,rmean=8)

```

```

Call: survfit(formula = Surv(months, event) ~ treat, data = ipass[ipass$egfr ==
1, ])

```

```

      n events rmean* se(rmean) median 0.95LCL 0.95UCL
treat=0 129   104   5.82     0.205   6.82    5.95    6.99
treat=1 132    91   6.54     0.179   8.48    6.83    9.68
* restricted mean with upper limit = 8

```

```

#RMST (and their differences) using rmst2 function
rmst2(ipass$months[ipass$egfr==1], ipass$event[ipass$egfr==1],
      ipass$treat[ipass$egfr==1], tau=8)

```

The truncation time: tau = 8 was specified.

Restricted Mean Survival Time (RMST) by arm

	Est.	se	lower .95	upper .95
RMST (arm=1)	6.541	0.179	6.190	6.891
RMST (arm=0)	5.818	0.205	5.416	6.220

Restricted Mean Time Lost (RMTL) by arm

	Est.	se	lower .95	upper .95
RMTL (arm=1)	1.459	0.179	1.109	1.810
RMTL (arm=0)	2.182	0.205	1.780	2.584

Between-group contrast

	Est.	lower .95	upper .95	p
RMST (arm=1)-(arm=0)	0.722	0.189	1.256	0.008
RMST (arm=1)/(arm=0)	1.124	1.030	1.227	0.009
RMTL (arm=1)/(arm=0)	0.669	0.494	0.906	0.009

```
#risk diffs at 8m
risk.0<-1-summary(km,times=c(8))$surv[1]
risk.1<-1-summary(km,times=c(8))$surv[2]

varrisk.0<-summary(km,times=c(8))$std.err[1]^2
varrisk.1<-summary(km,times=c(8))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0

lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)

risk.1-risk.0
```

```
[1] -0.2664999
```

```
lower.ci.riskdiff
```

```
[1] -0.3886012
```

```
upper.ci.riskdiff
```

```
[1] -0.1443986
```

```
2*pnorm((risk.1-risk.0)/sqrt(var.riskdiff))#p-value
```

```
[1] 1.886459e-05
```

```
#-----
#RMST at 16 months
print(km,rmean=16)
```

```
Call: survfit(formula = Surv(months, event) ~ treat, data = ipass[ipass$egfr ==
1, ])
```

```

      n events rmean* se(rmean) median 0.95LCL 0.95UCL
treat=0 129   104   6.71    0.333   6.82    5.95    6.99
treat=1 132    91   9.12    0.429   8.48    6.83    9.68
* restricted mean with upper limit = 16
```

```
#RMST (and their differences) using rmst2 function
rmst2(ipass$months[ipass$egfr==1], ipass$event[ipass$egfr==1],
      ipass$treat[ipass$egfr==1], tau=16)
```

The truncation time: tau = 16 was specified.

```
Restricted Mean Survival Time (RMST) by arm
      Est.      se lower .95 upper .95
RMST (arm=1) 9.121 0.429    8.279    9.963
RMST (arm=0) 6.714 0.333    6.062    7.367
```

```
Restricted Mean Time Lost (RMTL) by arm
      Est.      se lower .95 upper .95
RMTL (arm=1) 6.879 0.429    6.037    7.721
RMTL (arm=0) 9.286 0.333    8.633    9.938
```

```
Between-group contrast
      Est. lower .95 upper .95 p
RMST (arm=1)-(arm=0) 2.407    1.341    3.472 0
RMST (arm=1)/(arm=0) 1.358    1.188    1.553 0
RMTL (arm=1)/(arm=0) 0.741    0.643    0.853 0
```

```
#risk diffs at 16m
risk.0<-1-summary(km,times=c(16))$surv[1]
risk.1<-1-summary(km,times=c(16))$surv[2]

varrisk.0<-summary(km,times=c(16))$std.err[1]^2
varrisk.1<-summary(km,times=c(16))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0
```

```
lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)

risk.1-risk.0
```

```
[1] -0.1847429
```

```
lower.ci.riskdiff
```

```
[1] -0.2784014
```

```
upper.ci.riskdiff
```

```
[1] -0.0910845
```

```
2*pnorm((risk.1-risk.0)/sqrt(var.riskdiff))#p-value
```

```
[1] 0.0001105737
```

At 8 months:

- RMST difference=0.7 months (95% CI 0.2-1.3), p=0.008.
- Risk difference=-27% (14% to 39%), p<0.0001.

At 16 months:

- RMST difference=2.4 months (95% CI 1.3-3.5), p<0.0001
- Risk difference=-18% (-9% to -28%), p=0.0001

The risk difference provides stronger evidence of treatment benefit at 8 months, but this is because the time-frame is inappropriately short and has not allowed given enough time for the treatment benefit to accumulate (since there is an initial period where the survival curves do not diverge much). By 16 months there is stronger evidence of benefit using RMST.

Question 4 (optional)

Discuss in groups of 3 or 4 people:

- Which measure of treatment effect would you recommend for a new trial focusing only on patients with an EGFR positive mutation?
- After feedback from the medicine's regulator, you are told that your new trial must use restricted mean progression-free survival time as the primary measure of treatment benefit. How would you choose the milestone time for the statistical analysis?

SOLUTIONS

This question is open-ended with no single correct answer. Points to make:

- Unlike in the full IPASS study population where the treatment appears initially harmful, in this subgroup of patients we expect the treatment to be protective for all (or at least the majority of) follow-up. Then a hazard ratio from a Cox model may be a more reasonable choice than before. It may still be difficult to interpret however.
- An issue with using the risk difference here would be the choice of the milestone time to use. If it is chosen too late the majority of patients will die irrespective of treatment. If it is too early then the survival curves may not yet have diverged between the two groups and/or many events may be excluded from the analysis, potentially harming statistical power.
- RMST difference may be useful here and would be easy to interpret. We should choose a late milestone time so that we capture the full effect of treatment (which kicks in later), which is practical here in any case since most patients have complete follow-up (since the vast majority have an event).

Part B: Questions on the EXCEL data

EXCEL_IPD contains data approximating the EXCEL trial referenced in the lecture notes and includes the following variables:

Variable	Description
patientID	unique patient identifier
months	time from randomization in months
event	1=primary composite outcome occurs, 0=censored
treat	1=percutaneous coronary intervention (PCI), 0=coronary artery bypass graft (CABG)

Load the data:

```
excel<-read.table("datasets/EXCEL_IPD.csv",sep="," ,header=T)
```

Aims

Key learning objectives:

- Examine how summary measures evolve over multiple follow-up times.

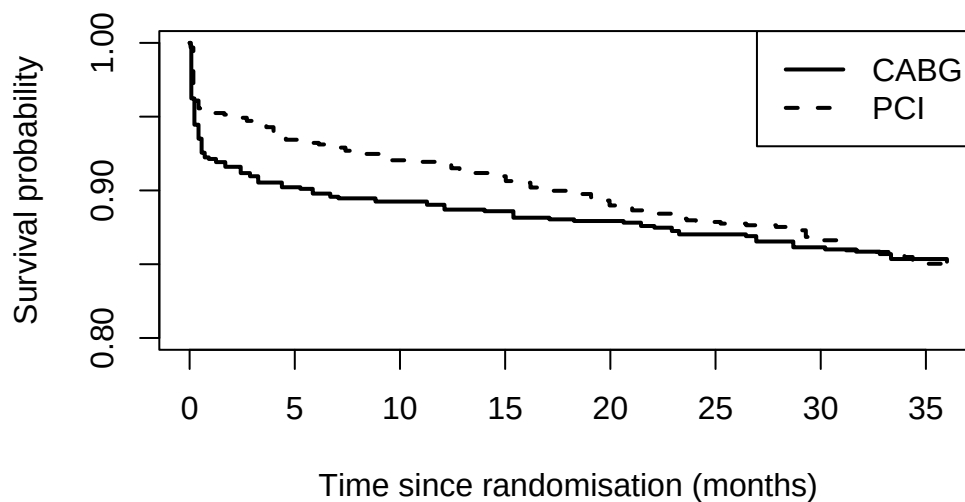
- Explore stability and interpretability of RMST vs risk difference.
- Use piecewise hazards models to handle potential non-proportional hazards
- Develop critical insight into composite outcomes and how different components contribute to overall trial results.

Question 1

Estimate the RMST and risk difference at 2 years, along with 95% confidence and p-values (adapt code provided in Part A). Comment on your findings.

SOLUTIONS

```
#Kaplan-Meier by treatment
km<-survfit(Surv(months,event)~treat,data=excel)
plot(km,ylab="Survival probability",
      xlab="Time since randomisation (months)",
      lwd=2,lty=c(1,2),ylim=c(0.8,1))
legend("topright",c("CABG","PCI"),
      lwd=2,lty=c(1,2))
```



```
#RMST at 24 months
print(km,rmean=24)
```

```
Call: survfit(formula = Surv(months, event) ~ treat, data = excel)
```

	n	events	rmean*	se(rmean)	median	0.95LCL	0.95UCL
treat=0	957	135	21.4	0.231	NA	NA	NA
treat=1	948	139	22.0	0.197	NA	NA	NA

* restricted mean with upper limit = 24

```
rmst2(excel$months, excel$event, excel$treat, tau=24)
```

The truncation time: tau = 24 was specified.

Restricted Mean Survival Time (RMST) by arm

	Est.	se	lower .95	upper .95
RMST (arm=1)	22.004	0.197	21.618	22.390
RMST (arm=0)	21.412	0.231	20.959	21.864

Restricted Mean Time Lost (RMTL) by arm

	Est.	se	lower .95	upper .95
RMTL (arm=1)	1.996	0.197	1.610	2.382
RMTL (arm=0)	2.588	0.231	2.136	3.041

Between-group contrast

	Est.	lower .95	upper .95	p
RMST (arm=1)-(arm=0)	0.592	-0.003	1.187	0.051
RMST (arm=1)/(arm=0)	1.028	1.000	1.056	0.052
RMTL (arm=1)/(arm=0)	0.771	0.594	1.001	0.051

```
#risks and risk diff at 24 months
risk.0<-1-summary(km,times=c(24))$surv[1]
risk.1<-1-summary(km,times=c(24))$surv[2]

#variance of risk diff, and 95% CI
varrisk.0<-summary(km,times=c(24))$std.err[1]^2
varrisk.1<-summary(km,times=c(24))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0

lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)
```

```
risk.1-risk.0
```

```
[1] -0.00954962
```

```
lower.ci.riskdiff
```

```
[1] -0.03954247
```

```
upper.ci.riskdiff
```

```
[1] 0.02044322
```

```
2*pnorm((risk.1-risk.0)/sqrt(var.riskdiff))#p-value
```

```
[1] 0.5325899
```

- At 2 years the RMST difference is 0.59 (95% CI 0.00 - 1.19) months, which is of borderline statistical significance ($p=0.051$).
- The risk difference is -1.0% (-4.0% to 2.0%) in favour of PCI, which is not statistically significant ($p=0.53$).
- The risk difference shows that there is not strong evidence that the risk at 2 years is greater on CABG. But the RMST is borderline statistically significant, because the gap between the survival curves is greater in the earlier part of the 2 year period, with the curve being lower for CABG, before they get closer together. The risk difference just looks at the difference between the curves *at* 2 years, whereas the RMST difference compares the areas under the curves *up to* 2 years.

Question 2

Estimate the risk difference and RMST difference at 2.5 and 3 years. What happens to the risk difference? What happens to the RMST difference?

SOLUTIONS


```
#rmst diffs at times 2.5, 3 yrs
rmst2(excel$months, excel$event, excel$treat, tau=30)
```

The truncation time: tau = 30 was specified.

Restricted Mean Survival Time (RMST) by arm

	Est.	se	lower .95	upper .95
RMST (arm=1)	27.257	0.256	26.756	27.758
RMST (arm=0)	26.612	0.294	26.037	27.188

Restricted Mean Time Lost (RMTL) by arm

	Est.	se	lower .95	upper .95
RMTL (arm=1)	2.743	0.256	2.242	3.244
RMTL (arm=0)	3.388	0.294	2.812	3.963

Between-group contrast

	Est.	lower .95	upper .95	p
RMST (arm=1)-(arm=0)	0.645	-0.118	1.408	0.098
RMST (arm=1)/(arm=0)	1.024	0.996	1.054	0.098
RMTL (arm=1)/(arm=0)	0.810	0.631	1.039	0.097

```
rmst2(excel$months, excel$event, excel$treat, tau=36)
```

The truncation time: tau = 36 was specified.

Restricted Mean Survival Time (RMST) by arm

	Est.	se	lower .95	upper .95
RMST (arm=1)	32.403	0.316	31.783	33.023
RMST (arm=0)	31.752	0.358	31.051	32.453

Restricted Mean Time Lost (RMTL) by arm

	Est.	se	lower .95	upper .95
RMTL (arm=1)	3.597	0.316	2.977	4.217
RMTL (arm=0)	4.248	0.358	3.547	4.949

Between-group contrast

	Est.	lower .95	upper .95	p
RMST (arm=1)-(arm=0)	0.651	-0.285	1.586	0.173
RMST (arm=1)/(arm=0)	1.020	0.991	1.051	0.174
RMTL (arm=1)/(arm=0)	0.847	0.667	1.075	0.172

```
#risks and risk diff at 2.5 yrs
risk.0<-1-summary(km,times=c(30))$surv[1]
risk.1<-1-summary(km,times=c(30))$surv[2]

varrisk.0<-summary(km,times=c(24))$std.err[1]^2
varrisk.1<-summary(km,times=c(24))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0

lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)

risk.1-risk.0
```

```
[1] -0.005907537
```

```
lower.ci.riskdiff
```

```
[1] -0.03590038
```

```
upper.ci.riskdiff
```

```
[1] 0.02408531
```

```
2*pnorm((risk.1-risk.0)/sqrt(var.riskdiff))#p-value
```

```
[1] 0.6994588
```

```
#risks and risk diff at 3 yrs
risk.0<-1-summary(km,times=c(36))$surv[1]
risk.1<-1-summary(km,times=c(36))$surv[2]

varrisk.0<-summary(km,times=c(24))$std.err[1]^2
```

```

varrisk.1<-summary(km,times=c(24))$std.err[2]^2

var.riskdiff<-varrisk.1+varrisk.0

lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)

risk.1-risk.0

```

```
[1] 0.00252326
```

```
lower.ci.riskdiff
```

```
[1] -0.02746958
```

```
upper.ci.riskdiff
```

```
[1] 0.03251611
```

```
2*pnorm(-(risk.1-risk.0)/sqrt(var.riskdiff))#p-value
```

```
[1] 0.8690287
```

- The risk difference decreases between 2 years and 2.5 years (to -0.6% in favour of PCI), and by 3 years is just in favour of CABG (0.3%), though is non-significant at all time points.
- The estimated RMST difference increases between 2 years and 2.5 years (RMST difference=0.65, 95% CI -0.12 to 1.41), but stays very similar at 3 years (RMST difference=0.65, 95% CI -0.29 to 1.59).
- Whereas the risk difference tends to decrease when more events are occurring in the intervention compared to the control arm, the RMST difference will continue to increase until the Kaplan Meier curves actually cross over.

	Estimate	95% CI	p-value
Risk difference (%)			
2 years	-0.9	(-4.0, 2.0)	0.53
2.5 years	-0.6	(-3.5, 2.4)	0.70
3 years	0.3	(-2.7, 3.3)	0.87

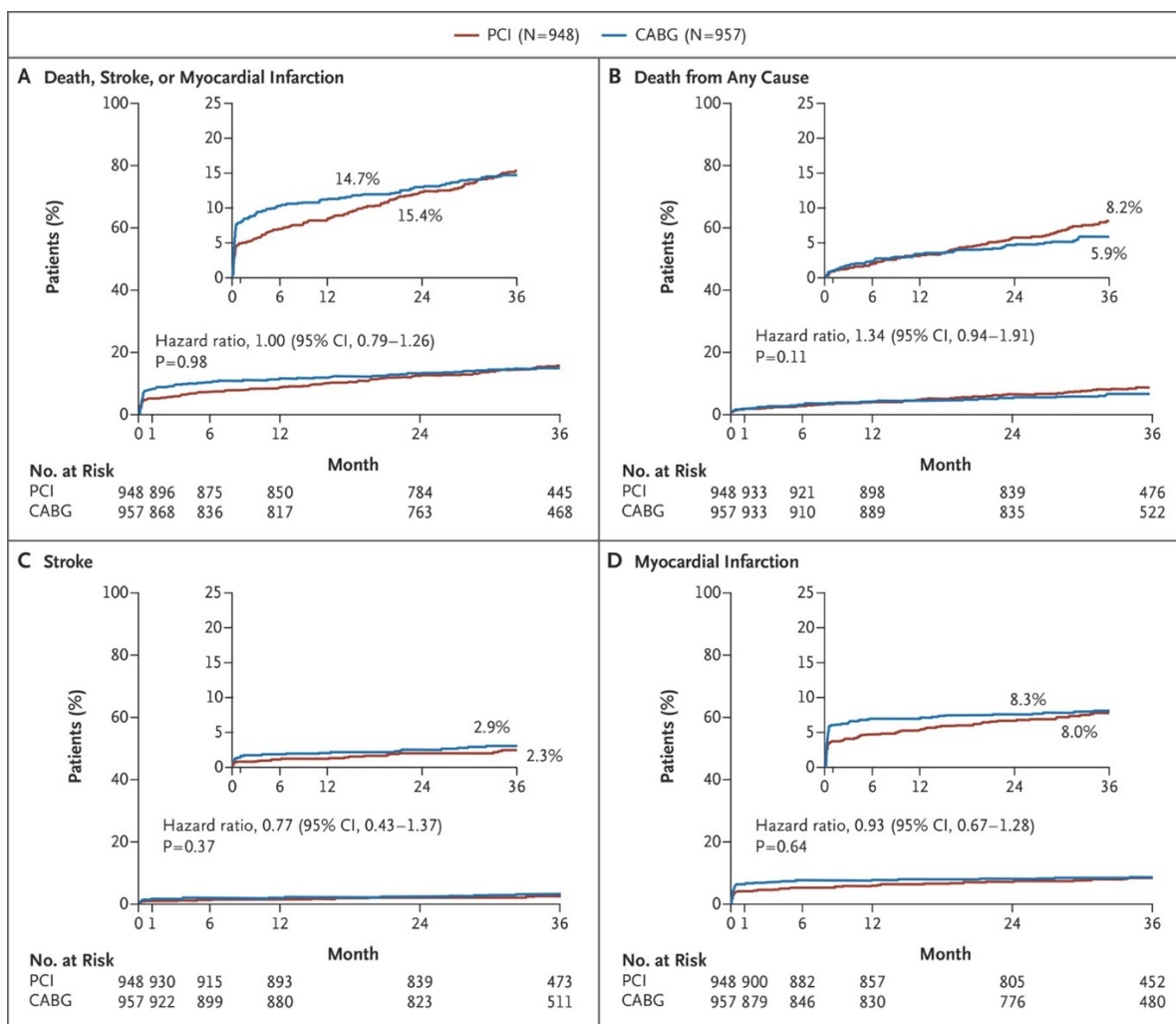
	Estimate	95% CI	p-value
RMST difference (months)			
2 years	0.59	(-0.003, 1.19)	0.051
2.5 years	0.65	(-0.12, 1.41)	0.098
3 years	0.65	(-0.29, 1.59)	0.173

Question 3

The figure below shows Kaplan-Meier estimates of the risks for the components of the primary outcome in the EXCEL trial are given below. What insights does this offer? Discuss in small groups how this may affect which treatment you would rather receive.

[Note that Kaplan-Meier estimates should not be used when there are competing events, as here, but that is what was presented for this study. For correct competing risks analysis see sessions 6 and 7 of the module].

```
knitr::include_graphics("09_practical_trials_fig1.png")
```



SOLUTIONS

The non-proportional hazards appears to be mainly driven by myocardial infarction, with an early benefit of PCI is by more later events. The most important outcome is 3 year all-cause death is higher on PCI (although not significantly). Overall the results when viewed separately by outcome are more favourable towards CABG than when the three endpoints are viewed as a composite.

Question 4 (optional)

Estimate the hazard ratio and its 95% confidence interval within each of the following periods after randomisation: (i) the first month; (ii) 1-12 months; (iii) 12 months-36 months. We do this using piecewise hazards models.

In R the data can be split into periods using the `SurvSplit` command used in Practical 5. Look at your data to see the new structure.

```
excel.split<-survSplit(Surv(months,event)~.,cut=c(1,12),
                      start="start.time",end="stop.time",
                      event="event",episode="period",data=excel)
```

Fit separate Cox proportional hazards models within each time period using (for example):

```
cox.mod.p1<-coxph(Surv(start.time,stop.time,event)~treat,
                  data=excel.split[excel.split$period==1,])
```

What additional insight does this analysis give you? What are the limitations?

SOLUTIONS

```
#cox model: allowing HR to vary in different time periods
#First split the data and generate the function of time we are interested in

excel.split<-survSplit(Surv(months,event)~.,cut=c(1,12),
                      start="start.time",end="stop.time",
                      event="event",episode="period",data=excel)

#One way of fitting the model(s)
cox.mod.p1<-coxph(Surv(start.time,stop.time,event)~treat,
                  data=excel.split[excel.split$period==1,])

cox.mod.p2<-coxph(Surv(start.time,stop.time,event)~treat,
                  data=excel.split[excel.split$period==2,])

cox.mod.p3<-coxph(Surv(start.time,stop.time,event)~treat,
                  data=excel.split[excel.split$period==3,])

#Another way of fitting the model
```

```
cox.mod2<-coxph(Surv(start.time,stop.time,event)~
                 treat*as.factor(period),
                 data=excel.split)
summary(cox.mod2)
```

Call:

```
coxph(formula = Surv(start.time, stop.time, event) ~ treat *
      as.factor(period), data = excel.split)
```

n= 5336, number of events= 274

	coef	exp(coef)	se(coef)	z	Pr(> z)	
treat	-0.5222	0.5932	0.1886	-2.770	0.005614	**
as.factor(period)2	NA	NA	0.0000	NA	NA	
as.factor(period)3	NA	NA	0.0000	NA	NA	
treat:as.factor(period)2	0.5480	1.7297	0.3198	1.713	0.086669	.
treat:as.factor(period)3	1.1076	3.0270	0.2896	3.824	0.000131	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
treat	0.5932	1.6858	0.4099	0.8584
as.factor(period)2	NA	NA	NA	NA
as.factor(period)3	NA	NA	NA	NA
treat:as.factor(period)2	1.7297	0.5781	0.9241	3.2376
treat:as.factor(period)3	3.0270	0.3304	1.7158	5.3401

Concordance= 0.552 (se = 0.015)

Likelihood ratio test= 15.41 on 3 df, p=0.001

Wald test = 14.77 on 3 df, p=0.002

Score (logrank) test = 15.15 on 3 df, p=0.002

0.5932*1.7297 #HR in period 2

[1] 1.026058

0.5932*3.0270 #HR in period 3

[1] 1.795616

```
#Another way of fitting this model
cox.mod3<-coxph(Surv(start.time,stop.time,event)~
  I(treat*I(period==1))+
  I(treat*I(period==2))+
  I(treat*I(period==3)),
  data=excel.split)
summary(cox.mod3)
```

Call:

```
coxph(formula = Surv(start.time, stop.time, event) ~ I(treat *
  I(period == 1)) + I(treat * I(period == 2)) + I(treat * I(period ==
  3)), data = excel.split)
```

n= 5336, number of events= 274

	coef	exp(coef)	se(coef)	z	Pr(> z)
I(treat * I(period == 1))	-0.52223	0.59319	0.18856	-2.770	0.00561 **
I(treat * I(period == 2))	0.02573	1.02606	0.25834	0.100	0.92067
I(treat * I(period == 3))	0.58534	1.79559	0.21985	2.662	0.00776 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
I(treat * I(period == 1))	0.5932	1.6858	0.4099	0.8584
I(treat * I(period == 2))	1.0261	0.9746	0.6184	1.7025
I(treat * I(period == 3))	1.7956	0.5569	1.1670	2.7628

Concordance= 0.552 (se = 0.015)

Likelihood ratio test= 15.41 on 3 df, p=0.001

Wald test = 14.77 on 3 df, p=0.002

Score (logrank) test = 15.15 on 3 df, p=0.002

Hazard ratios comparing PCI to CABG by time-frame are:

- Before 1 month: 0.59 (95% CI 0.41 to 0.86)
- 1 month-12 months 1.03 (95% CI 0.62 to 1.70)
- 12 months-36 months 1.80 (95% CI, 1.17 to 2.76)

PCI is protective in the short-term, similar from 1-month to 12 months and worse after 12 months. Based on these results, one may suspect that results with PCI may be worse in the period after the end of the trial. However, one should also bear in mind that only patients who do not have an early event feature in the analysis for later time periods, which relates to the challenges in interpreting hazard ratios that we have discussed.

Question 5 (discuss in groups of 3 or 4)

Suppose you are an investigator planning a follow-up study to EXCEL. You wish to use the same primary composite outcome of death, stroke or myocardial infarction. Consider how you would go about statistical planning, e.g. which method for the primary analysis, when would you evaluate the outcome, which methods would you use for secondary / supportive analyses?

SOLUTIONS

There is no single correct answer, but here are some considerations:

- It would be useful to have longer-term follow-up, to see whether the survival curves cross over after 3 years..
- Cox models assuming proportional hazards may not be a good choice here because of non-proportional hazards.
- Since most patients do not have an event, estimating a risk difference may be more appropriate than RMST.
- As supportive analyses, we could estimate RMST differences, or hazard ratios in different periods of follow-up.
- Given the results of EXCEL, viewing results by each component of the composite will remain useful.

Part C: Questions relating to ASCOT

ASCOT_IPD contains data approximating the data for cardiovascular death (a secondary outcome) in the blood pressure lowering arm of the Anglo Scandinavian Cardiovascular Outcomes Trial (ASCOT)

Variable	Description
patientID	unique patient identifier
years	time from randomization in years
event	1=cardiovascular death occurred, 0=censored
treat	1=Amlodipine; 0=Atenolol (two blood-pressure lowering medications)

Load the data:

```
ascot<-read.table("datasets/ASCOT_IPD.csv",sep=" ",header=T)
```

Learning objectives:

- Contrast RMST vs hazard ratio to summarise the treatment effect.
- Explore how the pattern of the treatment effect over time can influence statistical power and interpretability.

Question 1

Estimate the hazard ratio, RMST and risk difference at 5 years. Comment on your findings with respect to:

- which is most meaningful?
- which is most statistically powerful and why?
- how many events are excluded for the analysis based on RMST/risk difference?
- which treatment effect is superficially most impressive and why?

SOLUTIONS

```
#Hazard ratio
cox.mod<-coxph(Surv(years,event)~treat,data=ascot)
summary(cox.mod)
```

Call:

```
coxph(formula = Surv(years, event) ~ treat, data = ascot)
```

```
n= 19257, number of events= 535
```

```
      coef exp(coef) se(coef)      z Pr(>|z|)
treat -0.22300   0.80012  0.08829 -2.526   0.0115 *
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

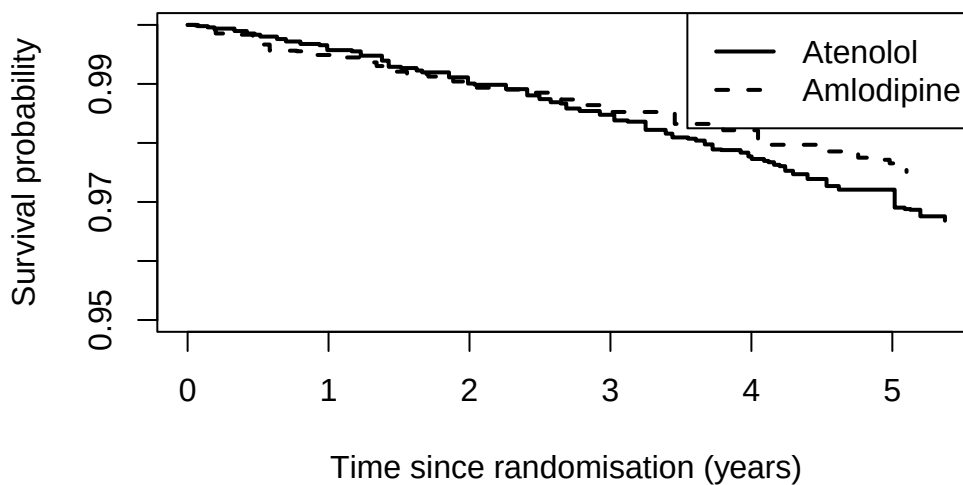
```
      exp(coef) exp(-coef) lower .95 upper .95
treat   0.8001      1.25    0.673   0.9513
```

```
Concordance= 0.527 (se = 0.011 )
```

```
Likelihood ratio test= 6.42 on 1 df,  p=0.01
```

Wald test = 6.38 on 1 df, p=0.01
Score (logrank) test = 6.41 on 1 df, p=0.01

```
#-----  
#Risk diff at 5 years  
  
km<-survfit(Surv(years,event)~treat,data=ascot)  
plot(km,ylab="Survival probability",  
      xlab="Time since randomisation (years)",  
      lwd=2,lty=c(1,2),ylim=c(0.95,1))  
legend("topright",c("Atenolol","Amlodipine"),  
      lwd=2,lty=c(1,2))
```



```
risk.0<-1-summary(km,times=c(5))$surv[1]  
risk.1<-1-summary(km,times=c(5))$surv[2]  
  
varrisk.0<-summary(km,times=c(5))$std.err[1]^2  
varrisk.1<-summary(km,times=c(5))$std.err[2]^2  
  
var.riskdiff<-varrisk.1+varrisk.0  
  
#95% confidence interval  
lower.ci.riskdiff<-(risk.1-risk.0)-1.96*sqrt(var.riskdiff)  
upper.ci.riskdiff<-(risk.1-risk.0)+1.96*sqrt(var.riskdiff)  
  
risk.1-risk.0
```

```
[1] -0.004454733
```

```
lower.ci.riskdiff
```

```
[1] -0.008994868
```

```
upper.ci.riskdiff
```

```
[1] 8.540098e-05
```

```
2*pnorm((risk.1-risk.0)/sqrt(var.riskdiff))#p-value
```

```
[1] 0.05446348
```

```
#-----  
#RMST at 5 years  
  
#in years  
rmst2(ascot$years, ascot$event, ascot$treat, tau=5)
```

The truncation time: tau = 5 was specified.

Restricted Mean Survival Time (RMST) by arm

	Est.	se	lower .95	upper .95
RMST (arm=1)	4.941	0.005	4.932	4.949
RMST (arm=0)	4.934	0.005	4.925	4.943

Restricted Mean Time Lost (RMTL) by arm

	Est.	se	lower .95	upper .95
RMTL (arm=1)	0.059	0.005	0.051	0.068
RMTL (arm=0)	0.066	0.005	0.057	0.075

Between-group contrast

	Est.	lower .95	upper .95	p
RMST (arm=1)-(arm=0)	0.006	-0.007	0.019	0.343
RMST (arm=1)/(arm=0)	1.001	0.999	1.004	0.343
RMTL (arm=1)/(arm=0)	0.907	0.740	1.111	0.344

```
#in days
rmst2(ascot$years*365.25, ascot$event, ascot$treat, tau=5*365.25)
```

The truncation time: tau = 1826.25 was specified.

Restricted Mean Survival Time (RMST) by arm

	Est.	se	lower .95	upper .95
RMST (arm=1)	1804.526	1.668	1801.257	1807.795
RMST (arm=0)	1802.292	1.661	1799.036	1805.548

Restricted Mean Time Lost (RMTL) by arm

	Est.	se	lower .95	upper .95
RMTL (arm=1)	21.724	1.668	18.455	24.993
RMTL (arm=0)	23.958	1.661	20.702	27.214

Between-group contrast

	Est.	lower .95	upper .95	p
RMST (arm=1)-(arm=0)	2.234	-2.380	6.848	0.343
RMST (arm=1)/(arm=0)	1.001	0.999	1.004	0.343
RMTL (arm=1)/(arm=0)	0.907	0.740	1.111	0.344

The results are as follows:

- Hazard ratio: 0.80 (0.67-0.95); p=0.012
- Risk difference: -0.4% (-0.9% to 0.0%), p=0.054
- RMST difference: 2.2 days (-2.4 to 6.8), p=0.34

Comments on the various analysis

- We would expect either the Cox model or the risk difference to be most powerful here since the benefit of treatment occurs late during follow-up.
- 55 events occur after 5 years and so are excluded using an analysis with RMST or risk difference when estimated non-parametrically like this), but included in an analysis using a Cox model. As there are 535 events before 5 years, this is quite a large number of events to exclude. This may partly explain why an analysis using a Cox model is more statistically powerful. Especially since such events were far more common on atenolol (43) than amlodipine (12)

- When event rates are fairly low, measures on the relative scale can seem more impressive, e.g. the hazard ratio of 0.80 is perhaps superficially more impressive than the risk difference of 0.4%. The RMST difference is an absolute measure of treatment benefit and is often superficially underwhelming. In this example the RMST of 2.2 days is also superficially unimpressive because:
 - i) The benefit of amlodipine does not occur until late, and so there is limited time for the RMST difference to emerge/accrue.
 - ii) RMST is highly dependent upon the time horizon used in the analysis. It is important to remember that much benefit of treatment may occur after the end of a trial, where because fewer patients have an event with amlodipine the treatment benefit will continue to accumulate. This can be visualized in the figure below which shows that in a hypothetical scenario where events were balanced between treatment arms (i.e. cumulative incidence curves run parallel) the RMST difference continues to accrue so that the benefit at 10 years is larger than at 5 year (note this figure uses the real ASCOT data, for the blood pressure lowering arm, and so the results at 5 years differ slightly from those in the approximated data used in the practical session).